RSBEV: Multiview Collaborative Segmentation of 3-D Remote Sensing Scenes With Bird's-Eye-View Representation

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***Abstract*— Perception of 3-D remote sensing scenes plays a crucial role in accurately recognizing and locating ground objects, as it enables a deeper understanding of complex environments by capturing scene geometry, object relationships, and occlusion patterns. Inspired by the powerful multisensor fusion capabilities in autonomous driving, we explore a new task in this article: given a set of multiview images of a 3-D remote sensing scene, we aim to obtain bird's-eye-view (BEV) scene information under the common view area in the world coordinate system. In this work, we focus on the task of semantic segmentation to demonstrate the feasibility of our approach and introduce a BEV modeling technique tailored for remote sensing scenes, which facilitates the projection of 3-D scene details from multiple perspective views onto a BEV. We then utilize a dual-encoder structure based on the vision transformer (VIT) architecture to extract relevant spatial information using self-attention mechanisms. Within the decoder, we employ a feature pyramid network (FPN) to integrate BEV patch encoding with spatial feature residuals, enabling fine-grained segmentation results at the original input resolution. Furthermore, we curated the LEVIR-MDS multidrone segmentation dataset, comprising scenes from ten community-level areas across three continents, totaling 243k images and their corresponding annotated BEV semantic maps, amounting to approximately 500 GB. This dataset serves as a robust benchmark to assess the effectiveness and generalization capability of our proposed method. To our knowledge, this is the first semantic segmentation dataset designed specifically for collaborative multidrone applications. We further show that our method achieves a 12% improvement in mean IoU (mIoU), reaching 69.73%, compared to a pure convolutional network model.**

***Index Terms*— Bird's-eye-view (BEV) representation, multiview collaborative segmentation, remote sensing, semantic segmentation.**

Received 17 October 2024; accepted 1 November 2024. Date of publication 8 November 2024; date of current version 2 December 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022ZD0160401, in part by the National Natural Science Foundation of China under Grant 62125102 and Grant 62471014, in part by the Beijing Natural Science Foundation under Grant JL23005, and in part by the Fundamental Research Funds for the Central Universities. *(Corresponding author: Zhenwei Shi.)*

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Digital Object Identifier 10.1109/TGRS.2024.3493963

I. Introduction

MULTIVIEW stereoscopic perception is vital in remote sensing tasks for applications such as disaster assessment [2], [3], resource monitoring [4], [5], and land surveying [6]. While drone-based remote sensing provides advantages over satellite imaging, including flexible path planning and fine-grained perception, most existing remote sensing approaches rely on pixel-based segmentation, overlooking the richness of multiview and 3-D information, as illustrated in Fig. 1(a) and (b).

With the advancement of convolutional neural networks (CNNs), the performance of semantic segmentation has significantly improved. Existing drone-based remote sensing semantic segmentation can be categorized into two types based on the input perspective: semantic segmentation for side-view perspectives [1], [8] and semantic segmentation for top-view perspectives [2], [3], [9]. Side-view perspectives encompass more spatial information and perception coverage. However, they face challenges such as object occlusion at different heights, requiring semantic segmentation in a pixel-to-pixel space after perspective transformation, making it difficult to map to world coordinates. Top-view perspectives largely resolve occlusion issues and allow direct target localization in world coordinates. However, their perception range is fixed, with a single viewpoint that limits capturing spatial scene information.

In recent years, within the realm of autonomous driving (e.g., Tesla [10]), pertinent methods have adeptly integrated multiview side-view sensor data by creating a bird's-eye-view (BEV) representation. This method involves projecting and integrating information captured by side-view sensors around the vehicle into a top-down view. The process facilitates a comprehensive depiction of surrounding scenes in the world coordinate system, providing an absolute scale and leveraging 3-D information perceived from side-view perspectives.

To project side-view information from the BEV perspective, methods can be categorized into two approaches. Some methods directly map side-view features to BEV space, such as [11], [12], [13], and [14], by extracting 2-D image features from side views, predicting depth distribution, and then projecting and integrating them into BEV space. Others like [15] and [16] use depth estimation to predict side-view depth maps and extract features in 3-D space. These methods share a common goal of recovering depth information lost in the 2-D imaging process of side views and associating it with

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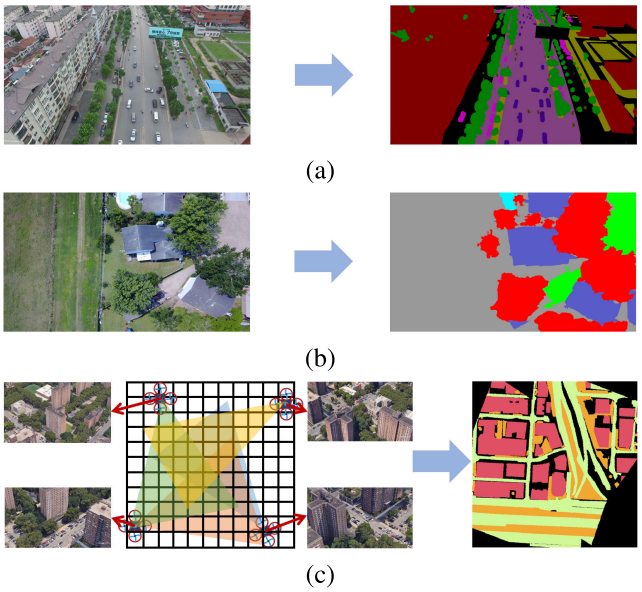


Fig. 1. Visual comparison between existing drone semantic segmentation framework and our proposed framework. (a) Captured from a side-view perspective and annotated using pixel-to-pixel labeling [1]. (b) Captured from a top-view perspective and annotated using pixel-to-pixel labeling [2]. (c) Our proposed framework, which learns scene 3-D information from multiple side-view perspectives and performs semantic segmentation on BEV at an absolute scale.

BEV space. Alternatively, some methods utilize camera models for inverse perspective mapping (IPM) directly in BEV space. For example, Roddick et al. [17] first utilized this method in 3-D object detection, assuming targets lie on a plane in 3-D space and querying point-by-point for corresponding features from various viewpoints. Furthermore, Lu et al. [18] and Pan et al. [19] built mappings between BEV space points and side-view images based on a variational encoder–decoder (VED) and multi-layer perceptron (MLP) architecture, while Li et al. [20] and Gong et al. [21] utilized transformers to directly construct BEV queries and employ cross-attention mechanisms to search for corresponding features in perspective images.

However, these methods build on assumptions made for autonomous driving tasks, where the objective is limited to perceiving the immediate surroundings of a vehicle. This allows for treating the ground as a nearly flat plane to facilitate straightforward projections between perspective views and the BEV. In drone scenarios, by contrast, the task generally involves a much broader scope of scene perception, rendering the simplistic flat plane assumption impractical. Moreover, unlike in autonomous driving, the extensive freedom in drone trajectories and the multiscale nature of scenes demand a specialized BEV-based framework designed for drone applications. Therefore, we introduce a robust network framework, remote sensing bird's-eye view (RSBEV), tailored for multiview collaborative segmentation. Accompanying this framework, we present a substantial dataset named LEVIR-MDS, which spans ten community-level scenes across three continents, encompassing 243k images with corresponding annotated BEV semantic maps, totaling approximately 500 GB.

In this article, we apply IPM within the BEV framework to comprehensively extract all relevant 2-D data and employ self-attention mechanisms to meticulously filter and ascertain accurate 3-D information. Initially, we model the ground height under a normal distribution assumption using a multiplane approach and extract BEV representation of the scene through weighted pooling. Subsequently, we propose a dual encoder structure based on the vision transformer (VIT) [22] architecture to filter out irrelevant spatial information using self-attention mechanisms. Finally, in the decoder, we design a feature pyramid network (FPN) [23] to connect BEV patch encoding with spatial feature residuals, enabling the acquisition of finely segmented BEV images at the input resolution.

The contributions of our work can be summarized as follows.

1) We propose a multiview joint perception framework designed for 3-D remote sensing scenes. Given a set of multiview images, our aim is to accurately derive BEV scene information within the shared viewing area of the world coordinate system, as illustrated in Fig. 1(c). Our method stands out by leveraging complex 3-D scene data and drone collaboration, ensuring precise perception of the environment and we utilize the task of semantic segmentation to validate the efficacy of our framework.
2) *Novel Algorithm (RSBEV):* We propose the RSBEV method to achieve multiview collaborative semantic segmentation. Initially, we project 3-D scene information from multiple perspectives into BEV space using BEV modeling. Additionally, we design a dual Encoder structure based on the VIT architecture, utilizing self-attention mechanisms to extract relevant spatial information. In the decoder phase, we employ an FPN to connect BEV patch encoding with spatial feature residuals, enabling finely segmented BEV results at the input resolution. These innovative algorithmic designs enable our method to effectively address multiview collaborative scene segmentation tasks with drones.
3) *New Dataset (LEVIR-MDS):* We curate a comprehensive multiview collaborative segmentation dataset, which includes scenes from ten community-level areas across three continents, featuring a total of 243k images and their corresponding annotated BEV semantic maps, totaling approximately 500 GB. This dataset provides essential experimental resources and serves as a significant benchmark in the drone remote sensing field. It not only validates the effectiveness of our proposed RSBEV method but also fosters extensive research and development in this domain.

The rest of this article is organized as follows. Section II reviews related work in BEV representation and semantic segmentation. Section III details our proposed method. Section IV presents the details of LEVIR-MDS. Section V provides experimental results demonstrating the effectiveness and rationale of our method. Finally, we conclude in Section VI.

II. Related Work

A. Vision-Centric BEV Representation

Vision-centric BEV representation, with its capacity to seamlessly integrate 3-D information at a real-world scale, has increasingly captured the attention of both industry and academia [24]. The earliest works in this area date back over 30 years, introducing methods that project side-view perspectives onto BEV under rigid plane assumptions [25]. With advancements in deep learning, several methods [7],

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[26], [27], [28] have utilized BEV representation to transform features from 2-D perspective pixel coordinates to 3-D world coordinates in a BEV.

To mitigate severe deformation and occlusion issues during rigid transformation processes, some methods have introduced generative adversarial network (GAN) [29] to compensate for errors in projection. For instance, big variational generative adversarial network (BigVGAN) [30] takes three views as inputs: perspective view, homography view, and BEV. It employs multiple submodels based on a multi-GAN architecture to address significant viewpoint differences and severe deformations between perspective views and BEV. MonoLayout [31] predicts BEV layouts, leveraging adversarial feature learning to hallucinate scene details for occluded image regions. This approach tackles the challenge of estimating layouts beyond visible information and compensates for 3-D information loss due to projection.

However, these direct transformation methods still suffer from incomplete projections due to occlusions and errors caused by the planar assumption. To address these fundamental issues, Philion and Fidler [11], Reading et al. [12], Wang et al. [15], and You et al. [16] leverage depth prediction or the probability distribution of depths from 2-D side-view images to determine the position of feature vectors in the BEV space. Nonetheless, directly predicting missing depth information increases model complexity and hinders convergence. Alternatively, categorical depth distribution network (CaDNN) [12] incorporates supervision using depth information obtained directly from lidar during depth estimation, leading to a more precise transformation from perspective view to BEV.

Meanwhile, some methods continue to pursue accurate transformation processes exclusively through visual cues. VED [18] employs learnable parameters to implicitly express the mapping from perspective view to BEV, utilizing a variational encoder–decoder and MLP architecture to extract semantic information in 3-D space directly from monocular images. Building on this approach, Pan et al. [19] and Gong et al. [21] integrate features from multiple perspective views into BEV space, aggregating spatial information from various viewpoints on the BEV feature map. While these methods effectively mitigate errors in BEV modeling of nonflat terrain such as mountains or buildings, they may encounter overfitting issues due to the network's requirement to fit additional scene-specific geometric details. Notably, hyperspectral feature transfer (HFT) [32] devised a hybrid model that combines geometric transformation features with camera parameters and learnable features, surpassing previous methods in performance. However, further enhancements are necessary to optimize performance when integrating multiview information [24].

The advent of VIT [22] has brought remarkable advancements in applying transformer-based models to computer vision tasks. Certain transformer-based approaches have surpassed MLP-based methods and consistently demonstrate strong performance in multiview fusion scenarios. In the field of autonomous driving, Li et al. [20], Gong et al. [21], and Chen et al. [33] leverage cross-attention mechanisms to effectively retrieve image features from perspective views to map onto the BEV. In such methods, the primary trade-off lies in the use of sparse queries [34], [35], [36], which sacrifices spatial geometric structure awareness, while dense queries [20], [37] require exceedingly high computational complexity.

This article addresses the characteristics of drones in large-scale scene perception. By employing IPM combined with a VIT-based architecture [22], we filter out inaccurately projected spatial information, achieving a balance between computational complexity and accuracy.

B. Semantic Segmentation Using Encoder–Decoder Architecture

The encoder–decoder architecture framework is widely employed in pixel-level segmentation tasks. In this framework, the encoder typically reduces the spatial scale of the input image and maps semantic information from the spatial domain to high-dimensional feature vectors. Meanwhile, the decoder performs upsampling to map these feature vectors back to category information. FPN [23] was first introduced using the Visual Geometry Group (VGG) network as its encoder, where the original fully connected layers were replaced with a decoder capable of restoring resolution. To ensure multiscale spatial information, feature maps of different scales from the encoder were directly fused with corresponding layers of the decoder, preserving spatial features at various resolutions. Subsequently, more convolutional network architectures such as residual network (ResNet) [38], high-resolution network (HRNet) [39], and residual network with split attention (ResNeSt) [40] have also been used as encoders. However, due to pooling operations in the encoder and continuous upsampling in the decoder, it remains challenging to learn high-frequency details in images. U-shaped network (U-Net) [41] and its subsequent variants [42], [43], [44], [45] introduced several shortcuts in the encoder–decoder structure, further preserving multiscale information, and found wide applications in image segmentation in medical and remote sensing domains [46]. A similar U-Net structure was adopted in [46], incorporating deconvolution layers in the decoder to extract roads in drone remote sensing scenes. Additionally, segmentation network (SegNet) [47] systematically decodes encoder features layer by layer, preserving indices in max pooling to consider contextual information and maintain overall object-level consistency during segmentation. More recent research, spatial–spectral masked autoencoder (MAE) [48], extends the encoder–decoder paradigm by focusing on masked image modeling, where certain portions of the input are masked, and the model is trained to reconstruct those hidden parts. This approach enhances the model's ability to capture spatial and spectral relationships, which is particularly valuable in remote sensing tasks [49]. MAE's masked reconstruction strategy leverages the rich spatial–spectral information in remote sensing data, contributing to improved scene classification and feature extraction.

The exceptional global modeling capabilities of Transformer [50] have been extended to encoder–decoder-based image segmentation frameworks in research. For example, segmentation transformer (SETR) [51] follows the paradigm of VIT [22] in the encoder, splitting the image into patches and leveraging transformers to learn global information. The output tokens are then reassembled into feature maps and processed with upsampling and convolution in the decoder. On the other hand, segmenter [52] is

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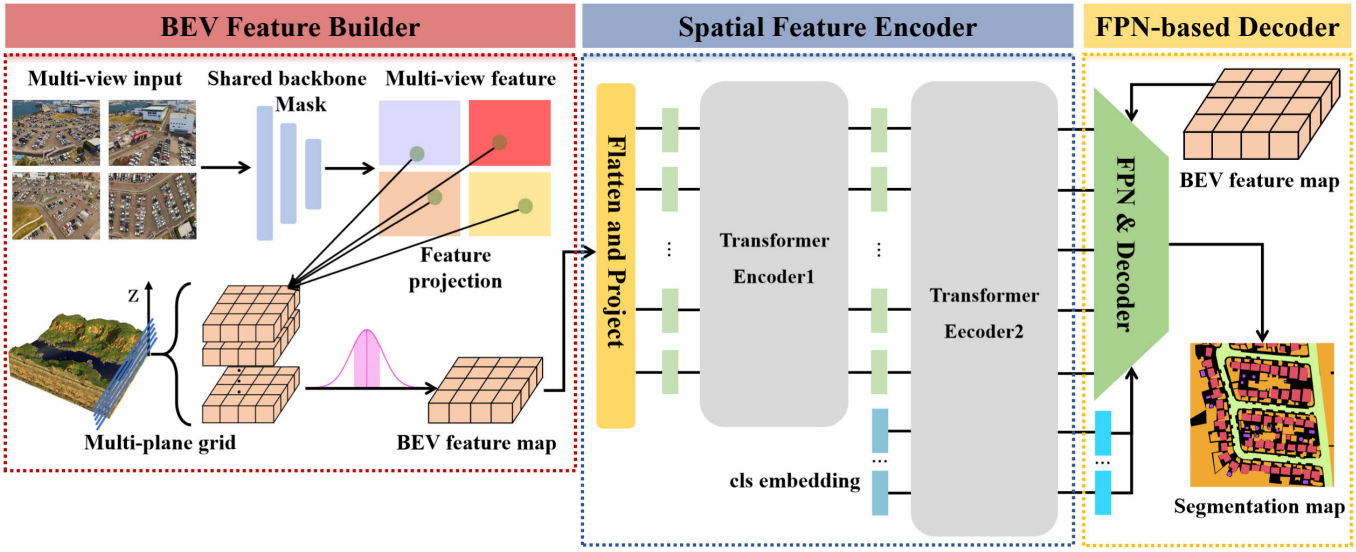


Fig. 2. Overview of our method. We take multiview images as input. In the BEV feature builder, multiview feature is projected onto a multiplane grid within the world coordinates, followed by a weighted pooling to obtain the BEV feature map. Subsequently, in the spatial feature encoder, the BEV feature map is used to learn spatial and category-related information. Finally, the FPN-based decoder predicts the segmentation map in BEV space.

built entirely on transformer-based encoder–decoder structures. In the decoding process, the authors replace convolutional architecture with a transformer that includes a category token and ultimately compute the similarity between output tokens and category tokens using dot products. Segformer [53] employs lightweight self-attention mechanisms and hierarchical feature maps in the encoder while leveraging MLP structures in the decoder to effectively reduce computational complexity. More recently, methods such as [54] have embedded Swin Transformer [55] into U-Net [41], resulting in improved segmentation accuracy for small-scale objects.

III. Proposed Method

The overall process of our RSBEV method is illustrated in the flowchart shown in Fig. 2. This method comprises three main components: 1) **BEV feature builder**, which projects features from input side-view perspectives to obtain BEV representations; 2) **Spatial feature encoder**, which filters out irrelevant spatial information from the BEV representations and models the correlation between features and categories; and 3) **FPN-based decoder**, which decodes the output features by integrating multiscale information to obtain category prediction results.

A. BEV Feature Builder

In this section, we efficiently project image features from multiple perspectives into BEV space. First, we extract features from multiperspective images using a shared backbone network. Subsequently, these features are explicitly projected onto a multiplane grid in the world coordinates using the camera models. This process allows us to obtain the BEV feature map through weighted pooling. The pseudocode illustrating the BEV Feature Builder process is provided in Algorithm 1.

We first process multiview images $\mathbf{I} \in \mathbb{R}^{N\times 3\times H\times W}$ and utilize their corresponding camera parameters $\mathbf{M} \in \mathbb{R}^{N\times 4\times 4}$ with a ResNet-like [38] network (NET) to extract image features. To effectively integrate spatial information, a depth coordinate map $\mathbf{Z} \in \mathbb{R}^{N\times 1\times H\times W}$, derived from $\mathbf{M}$, is introduced. This map, representing the depth for each viewpoint, quantifies the distance from the observed scene to the world coordinate plane at $z = 0$. It is concatenated with the extracted features, and a convolutional operation is applied to generate a per-pixel mask. This mask assesses the relative importance of image features at specific locations. The feature extraction process can be formalized as

$$\mathbf{F}_{\text{2-D}} = \text{Net}(\mathbf{I}) \tag{1}$$

$$\mathbf{mask} = \text{conv}(\text{concat}(\mathbf{F}_{\text{2-D}}, \mathbf{Z})) \tag{2}$$

$$\mathbf{F}_{\text{mask2D}} = \mathbf{F}_{\text{2-D}} \odot \text{sigmoid}(\mathbf{mask}) \tag{3}$$

where $\mathbf{F}_{\text{2-D}} \in \mathbb{R}^{N\times \text{feature_}h\times \text{feature_}w\times C}$ denotes the image features directly obtained from the feature extraction network, and $\mathbf{F}_{\text{mask2D}} \in \mathbb{R}^{N\times \text{feature_}h\times \text{feature_}w\times C}$ represents the image features weighted by the viewpoint-specific masks, where C is the dimension of the image features.

It is noteworthy that a significant challenge arises due to the unavailability of ground truth elevation values at arbitrary points within the scene. Directly projecting features onto the BEV plane from ground level often results in inaccurate feature representations. To address this issue, we model the ground height distribution within the scene as a Gaussian distribution $\mathcal{N}(\mu, \sigma)$, where μ represents the average scene height z_{world}, and we set $\sigma = 1.0$. Specifically, we discretize the height into a series of bins, dividing the space into $2L - 1$ parallel planes, equally spaced with elevations: $\mathbf{z} = \{z_1, z_2, \ldots, z_L, \ldots, z_{2L-2}, z_{2L-1}\}$, where $z_L = \mu$. In our experiments, we set $L = 4$ with a height spacing of 5 m, resulting in a total of seven planes. For each height z_i, a grid-shaped BEV map is created. Using the camera projection model, we search for 2-D image features that can be projected from each grid point onto the BEV plane. For a given BEV grid with coordinates (x, y, z), the projection process is mathematically described as follows:

$$\mathbf{p} = M_{\text{trans}}(\mathbf{M}, x, y, z) \tag{4}$$

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where $\mathbf{p} \in \mathbb{R}^{N\times 2}$ includes feature pixel coordinates of all cameras that can be projected from the coordinates (x, y, z). Further, we filter out points within the coordinate range of each camera's feature map, along with their corresponding features

$$\mathbf{p}_{\text{valid}} = \{\mathbf{p}_i \in \mathbf{p} | \text{check_bounds}(\mathbf{p}_i, H_{\text{feature}}, W_{\text{feature}})\} \tag{5}$$

$$\mathbf{F}_{\text{3-D}}[x, y, z] = \mathbf{F}_{\text{mask2D}}(\mathbf{p}_{\text{valid}}). \tag{6}$$

The function check_bounds verifies if the coordinates $\mathbf{p}_i$ fall within the feature map dimensions $[0, H_{\text{feature}}) \times [0, W_{\text{feature}})$. After filtering, $\mathbf{p}_{\text{valid}} \in \mathbb{R}^{n_{\text{hit}}\times 2}$ contains only those coordinates that meet these criteria, where $n_{\text{hit}} \in [0, N)$ denotes the count of viewpoints that can project their features within this coordinate range by the grid point. If $n_{\text{hit}} = 0$, then $\mathbf{F}_{\text{3-D}}[x, y, z]$ is a zero vector. Otherwise, $\mathbf{F}_{\text{3-D}}[x, y, z] \in \mathbb{R}^{n_{\text{hit}}\times C}$ represents the projected feature set corresponding to this BEV grid.

In the above process, the transformation $M_{\text{trans}}(\mathbf{M}, x, y, z)$ adheres to the pinhole camera model, where the 3-D point (x, y, z) in the world coordinate system is projected onto a 2-D pixel plane through the intrinsic and extrinsic camera parameters incorporated within the matrix $\mathbf{M}$. The projection mechanism is defined by the following equation:

$$\begin{pmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{1} \end{pmatrix} = \mathbf{K} \begin{pmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & \mathbf{1} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}. \tag{7}$$

In this equation, $(\mathbf{u}, \mathbf{v}) \in \mathbb{R}^{N\times 2}$ corresponds to the previously defined $\mathbf{p}$, $\mathbf{K}$ signifies the cameras' intrinsic matrix, $\mathbf{R}$ the rotation matrix, and $\mathbf{t}$ the translation vector. These components collectively compose the cameras' extrinsic matrix. The complete matrix $\mathbf{M}$ is thus formulated as the product of $\mathbf{K}$ and the concatenated extrinsic matrix $(\mathbf{R}\ \mathbf{t})$.

Once we have obtained the complete BEV multiplane features $\mathbf{F}_{\text{3-D}}(x, y, z) \in \mathbb{R}^{n_{\text{hit}}\times C}$, we compute the averaged features for each grid point in the BEV map as follows:

$$\mathbf{F}[x, y, z] = \begin{cases} \dfrac{1}{n_{\text{hit}}} \displaystyle\sum_{i=1}^{n_{\text{hit}}} \mathbf{F}_{\text{3-D}}[x, y, z][i], & \text{if } n_{\text{hit}} > 0 \\ \mathbf{0}, & \text{if } n_{\text{hit}} = 0 \end{cases} \tag{8}$$

where $\mathbf{F} \in \mathbb{R}^{\text{bev_}h\times \text{bev_}w\times(2L-1)\times C}$ represents the BEV map across different layers, providing a comprehensive representation of the scene at various heights.

Finally, we employ weighted pooling to aggregate $\mathbf{F}$ according to the discrete Gaussian distribution probabilities, yielding the final BEV feature map $\mathbf{B} \in \mathbb{R}^{\text{bev_}h\times \text{bev_}w\times C}$

$$\mathbf{B} = \sum_{i=0}^{2L-1} \frac{1}{\sigma\sqrt{2\pi}} \exp -\frac{(z_i - z_L)^2}{2\sigma^2} \cdot \mathbf{F}[:, :, i, :]. \tag{9}$$

Algorithm 1 BEV Feature Building Process. The Symbols Used in This Pseudocode Follow the Same Naming Conventions as in the Previous Formulas

Input: Multiview images $\mathbf{I}$, Camera parameters $\mathbf{M}$
Input: BEV grid dimensions (bev_h, bev_w, L)
Output: Final BEV feature map $\mathbf{B}$

Step 1: Extract multiview features.
mask $\leftarrow$ `conv(concat(Net(`$\mathbf{I}$`), `$\mathbf{Z}$`))`
// $\mathbf{Z}$: Depth map computed from $\mathbf{M}$
$\mathbf{F}_{\text{mask2D}} \leftarrow$ `Net(`$\mathbf{I}$`)` $\odot$ `sigmoid(`**mask**`)`

Step 2: For each grid point in the BEV map, calculate the projected feature.
for each grid point (x, y, z) in BEV grid **do**
  $\mathbf{p} \leftarrow M_{\text{trans}}(\mathbf{M}, x, y, z)$
  $\mathbf{p}_{\text{valid}} \leftarrow$ `check_bounds(`$\mathbf{p}$, $\mathbf{F}_{\text{mask2D}}$`.shape())`
  if $\mathbf{p}_{\text{valid}} \neq \emptyset$ **then**
    $\mathbf{F}_{3D}(x, y, z) \leftarrow \mathbf{F}_{mask2D}(\mathbf{p}_{\text{valid}})$
  else
    $\mathbf{F}_{3D}(x, y, z) \leftarrow \mathbf{0}$
  end if
end for

Step 3: Feature fusion and weighted pooling.
for each grid point (x, y, z) in BEV grid **do**
  if $n_{\text{hit}} > 0$ **then**
    $\mathbf{F}[x, y, z] \leftarrow \frac{1}{n_{\text{hit}}} \sum_{i=1}^{n_{\text{hit}}} \mathbf{F}_{3D}[x, y, z][i]$
  else
    $\mathbf{F}[x, y, z] \leftarrow \mathbf{0}$
  end if
end for
$\mathbf{B} \leftarrow$ `WeightedPooling(`$\mathbf{F}$, Gaussian weights`)`
return $\mathbf{B}$

B. Spatial Feature Encoder

We observe that the BEV feature map $\mathbf{B}$ obtained through pooling contains overlapping features from different elevations. Consequently, directly applying convolutional architectures for feature encoding and decoding would struggle to capture effective contextual information due to the presence of excessive invalid information in the BEV feature map. Subsequent experiments also validate the presence of severe overfitting under such conditions. To address this issue, we propose a transformer-based architecture inspired by VIT [22], which enables global modeling of spatial information through self-attention mechanisms. The entire process is divided into two encoders, Encoder1 and Encoder2, utilizing self-attention modules with identical structures. Furthermore, considering that the pixel-wise operations during the projection of image features may negate the regularization effect induced by data augmentation in subsequent encoders, we incorporate additional data augmentation on the BEV feature map $\mathbf{B}$ following the approach proposed in bird's-eye view detection (BEVDet) [13], involving operations such as rotation, cropping, resizing, and flipping.

In Encoder1, we divide the BEV map into a series of 8×8 patches, which are flattened and fed into a linear layer to obtain BEV feature tokens. These tokens, along with positional embeddings, are then input to Encoder1, comprising N_1 self-attention modules aimed at filtering out irrelevant spatial information. The output dimensions remain the same as the input BEV patch encoding. Subsequently, these encodings are combined with category embeddings and fed into Encoder2, which consists of N_2 self-attention modules designed to learn the dependencies between BEV feature encoding and category

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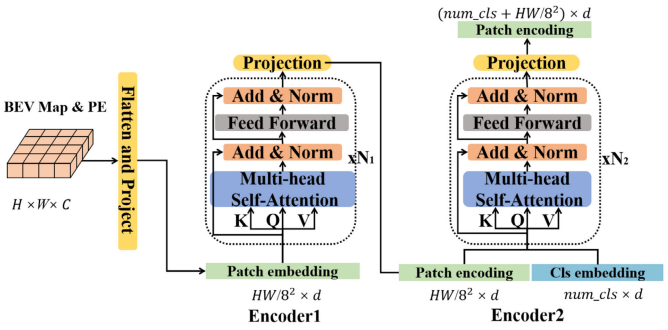


Fig. 3. Network architecture of the spatial feature encoder. In Encoder1, the network receives patch embedding with positional embedding and utilizes self-attention mechanisms to filter out irrelevant spatial information in the input. Subsequently, the output is concatenated with class embedding and fed into Encoder2 to learn the dependencies between features and class information.

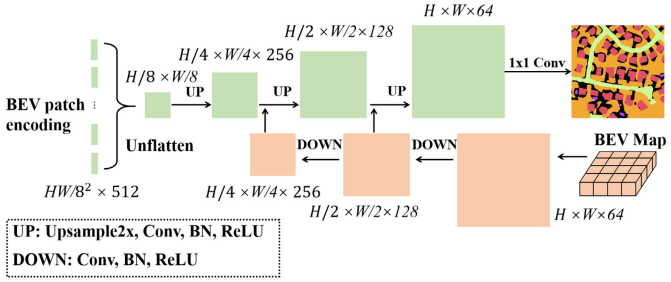


Fig. 4. Decoder network architecture diagram. This diagram illustrates how the feature sequence processed by the encoder is rearranged into a 2-D feature map and subsequently refined through three upsampling modules to produce the final segmentation results. During the upsampling process, feature maps of various resolutions are fused with the downsampled BEV feature map to enhance segmentation performance.

information. Finally, Encoder2 outputs the BEV patch encoding corresponding to each patch.

As shown in Fig. 3, the same self-attention operation is utilized in different encoders. **Q**, **K**, **V** denote query, key, and value, and input tokens (**Q**, **K**, V) are linearly projected using embedding tokens or encoding tokens T, represented as

$$\mathbf{Q} = \mathbf{T}\mathbf{W}^q, \quad \mathbf{K} = \mathbf{T}\mathbf{W}^k, \quad \mathbf{V} = \mathbf{T}\mathbf{W}^v \tag{10}$$

where $\mathbf{W}^q$, $\mathbf{W}^k$, and $\mathbf{W}^v$ are the learnable parameters of the linear projection layer. In Encoder1, they have dimensions $HW/8^2 \times d$, and in Encoder2, their dimensions are $(HW/8^2 + \text{num_cls}) \times d$. Moreover, num_cls represents the number of classes and d denotes the feature dimension of the tokens. After obtaining **Q**, **K**, and **V**, one attention head can be expressed as

$$\text{SelfAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}. \tag{11}$$

C. FPN-Based Decoder

Although our encoder contains intermediate features with precise spatial relations, the network struggles to capture information about small-scale objects. This is because each encoding vector at the output of the spatial feature encoder describes only the category information within each patch. Directly upsampling and computing the inner product with the class encoding, as done in [52], results in the loss of small-scale information at this stage. To address this issue, we introduce FPN [23] into the segmentation output head to fuse multiscale information. By employing both upscaling UP modules and downscaling DOWN modules, we fuse the BEV map before the spatial feature encoder with the feature maps concatenated with BEV patch encodings, enabling multiscale fusion. This ultimately yields category segmentation results in the BEV space at the original scale. The process is illustrated in Fig. 4.

D. Loss Function

Our loss function consists of three components: weighted cross-entropy loss L_{ce}, dice loss [56] L_{dice}, and inter-class diversity loss L_{cls}. Assuming C is the number of classes, N is the number of samples, y_{ij} denotes the ground truth label (0 or 1) of a sample i belonging to class j, and p_{ij} is the probability predicted by the model that samples i belongs to class j.

L_{ce} can be represented as follows:

$$L_{\text{ce}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} w_j \cdot y_{ij} \cdot \log\left(p_{ij}\right) \tag{12}$$

where w_j is the weight assigned to class j, which is set inversely proportional to the pixel count of each class to further balance the class distribution in the samples.

L_{dice} can be represented as follows:

$$L_{\text{dice}} = 1 - \frac{1}{C}\sum_{j=1}^{C}\frac{2 \cdot \sum_{i=1}^{N} p_{ij} \cdot y_{ij}}{\sum_{i=1}^{N} p_{ij} + \sum_{i=1}^{N} y_{ij}}. \tag{13}$$

L_{cls} can be represented as follows:

$$L_{\text{cls}} = -\frac{1}{C(C-1)}\sum_{i=1}^{C}\sum_{j=1, j\neq i}^{C}\left|\frac{\mathbf{u}_i}{\|\mathbf{u}_j\|} - \frac{\mathbf{u}_i}{\|\mathbf{u}_j\|}\right| \tag{14}$$

where $\mathbf{u}_i$ and $\mathbf{u}_j$ are both class encoding output by the encoder. Minimizing this ensures that features from different classes have distinct separability in the feature space, thereby enhancing the ability to distinguish between different classes. The final loss function can be represented as

$$L = L_{\text{ce}} + L_{\text{dice}} + \lambda L_{\text{cls}}. \tag{15}$$

The weight of L_{cls}, denoted by λ, is set to 0.1 in this case.

E. Network Details

1) BEV Feature Builder: As shown in Table I, we employ a ResNet-50-like [38] backbone network to extract image features. The average pooling and fully connected layers are removed and replaced with an upsampling network structured as an FPN [23], with corresponding feature maps being skip-connected. The output image feature size is half of the input image size, with feature channels set to $C = 64$.

The BEV feature map **B** was set to have dimensions $512 \times 512 \times 64$, corresponding to a spatial resolution of 0.8×0.8 m, effectively covering a perception range of 409.6×409.6 m. In the multiplane projection part of the BEV feature builder, we set $L = 4$ with a height spacing

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TABLE I
DETAILED CONFIGURATION OF OUR SHARED BACKBONE. HERE, CONV-DOWN REPRESENTS A CONVOLUTIONAL LAYER FOLLOWED BY BATCH NORMALIZATION, RECTIFIED LINEAR UNIT (RELU), AND MAX POOLING; CONV-UP REPRESENTS A CONVOLUTIONAL LAYER FOLLOWED BY RELU AND UPSAMPLING LAYER

Layer	Config	Input Dim	Output Dim	Resolution
Layer0	Conv-Down	3	64	1/4
Layer1	Resnet.layer1	64	256	1/4
Layer2	Resnet.layer2	256	512	1/8
Layer3	Resnet.layer3	512	1024	1/16
Layer4	Resnet.layer4	1024	2048	1/32
Conv_FPN1	Conv-UP+Layer4	2048	1024	1/16
Conv_FPN2	Conv-UP+Layer3	1024	512	1/8
Conv_FPN3	Conv-UP+Layer2	512	256	1/4
Conv_FPN4	Conv-UP+Layer1	256	64	1/2
Conv_mask	Concat-Conv	64+1	1	1/2
Scalar_product	Conv_mask $\odot$ Conv_FPN4	64&1	64	1/2

of 5 m, resulting in a total of seven planes, and the perception of information within a 30-m height range.

2) Transformer Encoder: We divide the input image into a series of 8×8 patches and then project each patch into a 256-D feature vector using a linear transformation (with an 8×8 convolutional kernel and a stride of 8). After incorporating positional embedding, these feature vectors serve as the input to the transformer. Additionally, in Table VII, we obtain models with different parameter counts by modifying the settings in the encoder. For instance, in RSBEV-base, we set the number of layers in both Encoder1 and Encoder2 to 6, the number of attention heads to 4, and the dimension of the feed forward layer to 512.

3) Segmentation Head: We reorganize the patch encodings outputted by the encoder into a shape of $(H/8) \times (W/8) \times 256$. Then, leveraging the FPN [23] structure depicted in Fig. 4, we progressively fuse BEV features with patch encodings at different scales. Finally, employing a 1×1 convolution, we predict categories at the original scale of the BEV feature map.

IV. DATASET

Multiview drone datasets have made significant progress in the field of object tracking [57], [58], yet there are still substantial gaps in the segmentation domain. In the realm of autonomous driving, several large-scale datasets such as nuScenes [59], KITTI [60], and the Waymo Open Dataset (WOD) [61] have been established, providing a wealth of data for various tasks. To validate the effectiveness of RSBEV, we drew inspiration from both dataset creation methods in the drone domain and dataset formats prevalent in the autonomous driving domain and proposed a comprehensive dataset named **LEVIR-MDS**, specifically designed for multiview collaborative segmentation.

This dataset leverages real-world 3-D scenes generated from Google Earth Studio [62], comprising ten extensive community-level scenes across three continents. Each scene is captured from four unique drone viewpoints, resulting in a comprehensive collection of 5800 frames per viewpoint. These frames have been rigorously annotated to categorize six distinct features: building, road, vegetation, water, boat, and others, demonstrating the substantial effort involved in dataset preparation. Representative images from the dataset are presented in Fig. 5, while the corresponding annotated BEV images are illustrated in Fig. 6.

Our LEVIR-MDS offers a robust foundation for advancing research in multiview collaborative segmentation, bridging the existing gaps in the current drone segmentation datasets. By leveraging the high-resolution and detailed nature of the simulated 3-D scenes, researchers can develop and validate new algorithms and methodologies aimed at enhancing segmentation accuracy and efficiency in multidrone systems.

A. Trajectory Setup

For each scene measuring approximately 4×4 km, the area is typically divided into three rows and five columns, and scanned along a serpentine trajectory through these subregions as illustrated in Fig. 7. During the scanning process, the camera centers of the four trajectory cameras in each frame converge on the same point, with the cameras facing forward, backward, leftward, and rightward, respectively. In our dataset, the BEV center position for each frame is determined by a meticulously designed set of trajectories, ensuring that the central position is as much as possible within the common view area of the four viewpoints.

Scanning is performed along the row direction with a 500-frame interval between two adjacent rows (three rows in total) and along the column direction with a 200-frame interval between two adjacent columns (five columns in total), all at a frame rate of 24 frames/s. These speeds were chosen to balance between detailed temporal resolution and practical data collection rates. This results in a sequence of 4×5800 temporal images for each scene, where each image measures 1600×900 pixels. The horizontal field of view (HFOV) for each image is 20°.

The height and tilt angle of the drones vary within specific ranges to simulate realistic flight conditions. Specifically, the height ranges from 100 to 500 m, and the tilt angle varies between 25° and 50° to capture different perspectives and enhance the dataset's diversity.

B. Data Pairing

For scene i, we only annotate to obtain the BEV ground-truth map Mask_i. In frame t, the visible area of the ground-truth map P^t in this scene is defined as

$$P^t = \bigcup_{n=1}^{N} \left(\bigcup_{p \in \text{Mask}_i} p = (u, v) \middle| M_{\text{trans}}(p) \in \text{grid}(I_n) \right) \quad (16)$$

where N represents the number of cameras, $p = (u, v)$ denotes the pixel coordinates in Mask_i, $\text{grid}(I_n)$ denotes the tensor of pixel coordinates in the nth viewpoint, and $M_{\text{trans}}(\cdot)$ denotes the projection from BEV to side-view transformation using the pinhole camera model.

Thus, we obtain the data pairs of the side-view images $\mathbf{I}_i^t$ and the corresponding ground truth mask $\text{mask}_i^t = \text{Mask}_i(P^t)$ in the BEV space, as illustrated in Fig. 1(c).

C. Dataset Format

The dataset structure is inspired by the nuScenes [59] dataset. It organizes data in directories named "sample," "sweep," and "mask," with metadata stored in "data_meta.json."

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Fig. 5. Sample images of the first four scenes from the dataset.

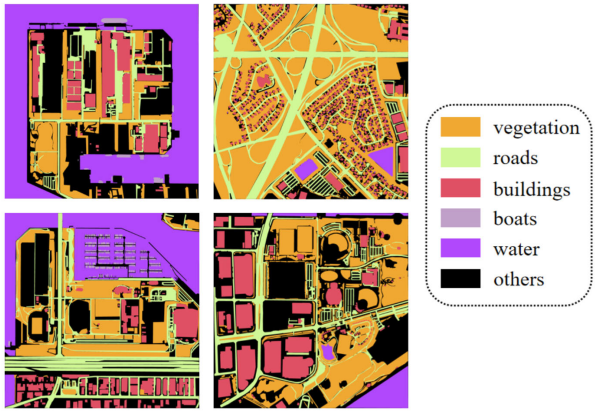


Fig. 6. Annotated BEV map of the first four scenes from the dataset.

1) Directory Structure: LEVIR-MDS includes the following subdirectories.

1) *Sample:* Stores keyframe images from different side-view perspectives in JPEG format. Each keyframe represents significant moments in each trajectory and is stored with a corresponding timestamp.
2) *Sweep:* Contains time-series sequences of images captured between keyframes from different side-view perspectives, also in JPEG format, providing a continuous view of each trajectory.
3) *Mask:* Contains the ground truth images for each scene, annotated with six categories. The annotated ground-truth images are stored in PNG format, along with the corresponding camera parameter information.
4) *data_meta.json:* Contains a dictionary that records the paths of images and corresponding camera parameters for all trajectories in each scene. The dictionary structure allows easy access to the dataset's metadata.

2) Statistical Properties: Key statistical properties of the dataset include the following.

1) *Scenes Information:* The dataset comprises a total of ten scenes sourced from three different continents. Specific details about each scene, including coordinates, area, average height, number of frames, and main categories, are provided in Table II.
2) *Number of Images:* Each scene contains a sequence of 4×5800 to 4×8500 frames captured from four distinct drone viewpoints, culminating in a total of 242 800 images.
3) *Image Resolution:* Each image has a resolution of 1600×900 pixels.
4) *Categories:* The ground-truth annotations include six categories: building, road, vegetation, water, boat, and other. The distribution of these categories across the dataset is illustrated in Fig. 8.
5) *Altitude Distribution:* Each scene's four trajectories are meticulously designed with a specific altitude distribution, ensuring a consistent shared viewing area among different drone perspectives throughout their motion. This design also facilitates the occurrence of image occlusions among various viewpoints. Altitude curves in the first scene are illustrated in Fig. 9.

V. Experiment

A. Experimental Setup

1) Implementation Details: We perform training on eight scenes, validation on one scene, and testing on a separate scene. Our network is deployed under the PyTorch framework.

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Fig. 7. Trajectories of the first four scenes and forward-looking images at different times. The leftmost images show the camera trajectories (white curves) from a forward-looking perspective for four scenes. White dots indicate the key points selected during the trajectory design, with camera poses for the remaining frames obtained using cubic spline interpolation for a smooth trajectory. The orange, green, and red dots represent the camera's position at three specific moments, corresponding to the images shown in the right three columns.

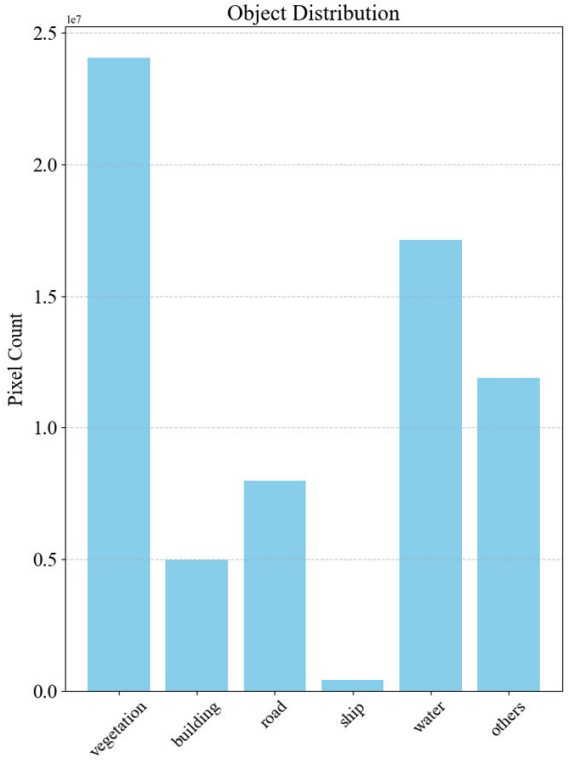


Fig. 8. Pixel count distribution of annotated categories.

We utilize a single NVIDIA GeForce RTX 4090 24-GB GPU. We employ the Adam optimizer with a learning rate of $5e^{-4}$ and a weight decay of $1e^{-8}$. The learning rate scheduler is configured to decay by 0.99 after each epoch. A batch size of 1 is utilized, and the maximum number of epochs is set to 100.

2) Evaluation Metrics: We employ intersection over union (IoU) to assess the classification performance of each category and use the average F1 score to evaluate the overall segmentation effectiveness of the network.

The IoU for each category i is calculated using the formula

$$\mathrm{IoU}_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{17}$$

where TP_i represents the true positives, FP_i represents the false positives, and FN_i represents the false negatives for category i.

The average F1 score is calculated using the formula

$$\mathrm{F1}_{\mathrm{avg}} = \frac{1}{C}\sum_{i=1}^{C}\frac{2\cdot \mathrm{Precision}_i \cdot \mathrm{Recall}_i}{\mathrm{Precision}_i + \mathrm{Recall}_i} \tag{18}$$

where $\mathrm{Precision}_i$ is given by $(\mathrm{TP}_i/(\mathrm{TP}_i + \mathrm{FP}_i))$ and Recall_i is given by $(\mathrm{TP}_i/(\mathrm{TP}_i + \mathrm{FN}_i))$.

B. Comparative Experiments

To further strengthen the evidence supporting the effectiveness of our proposed method and the newly introduced dataset, LEVIR-MDS, we conducted comparative experiments using two representative camera-based methods: camera to bird's-eye view (Cam2BEV) [63] and lift-splat-shoot (LSS) [11]. We also designed a purely convolutional network for comparison with our method.

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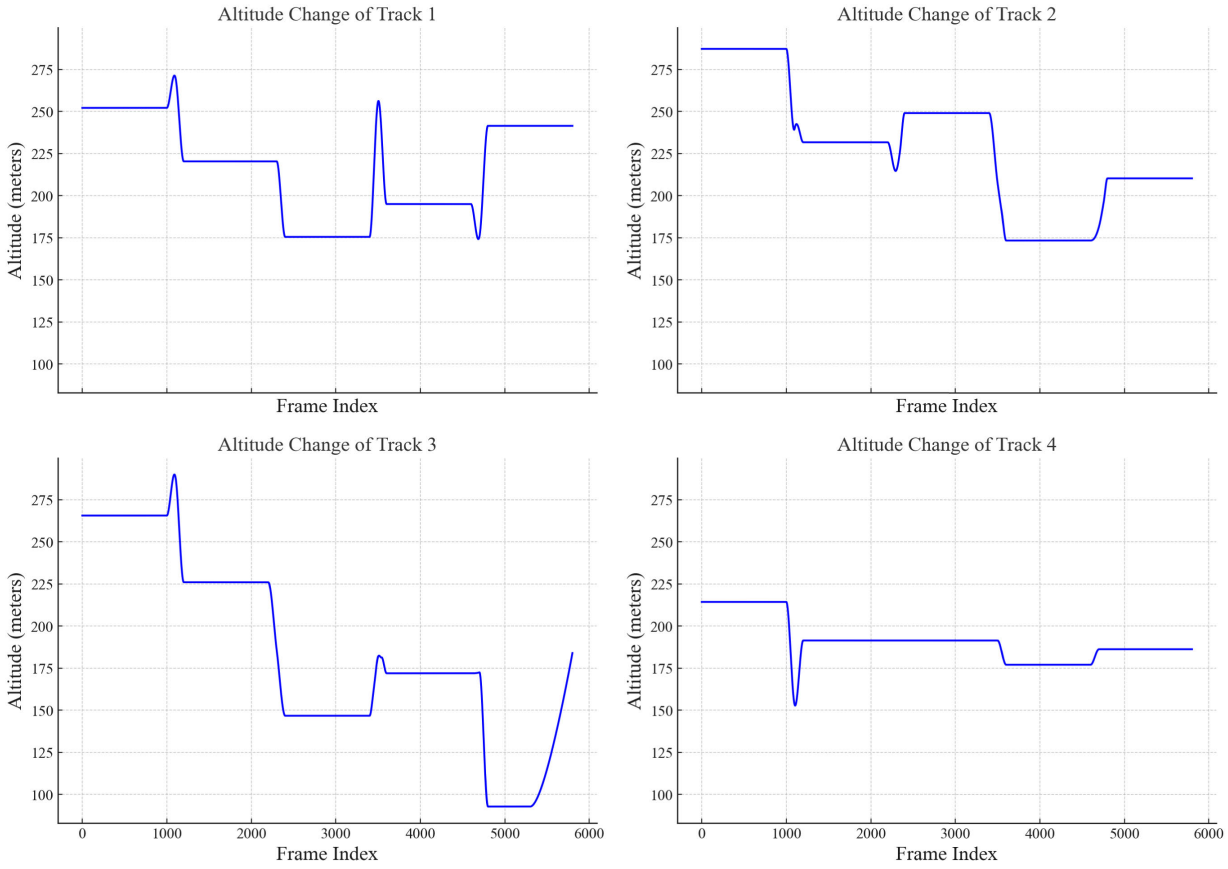


Fig. 9. Altitude curves for the four tracks in the first scene.

TABLE II
DETAILED INFORMATION OF EACH SCENE

Scene	latitude&longitude	Area (km^2)	Avg Height (m)	Frames	Main Categories	Region
Scene 1	47.62, -122.34	11	205.69	5800	Building, Ship	Washington, USA
Scene 2	40.79, -73.96	12	238.31	5800	Vegetation, Building	New York, USA
Scene 3	34.67, 135.40	14	297.13	5800	Vegetation, Building	Tokyo, Japan
Scene 4	35.65, 139.83	18	396.75	5800	Vegetation, Ship	Yokohama, Japan
Scene 5	35.45, 139.96	10	170.13	5800	Building	Yokohama, Japan
Scene 6	40.79, -74.12	14	244.24	5800	Vegetation, Water	New York, USA
Scene 7	40.80, -73.49	16	235.01	5800	Building, Roads	New York, USA
Scene 8	40.79, -74.01	15	245.62	5800	Vegetation, Water	Brooklyn, USA
Scene 9	47.46, 8.58	8	112.28	8500	Vegetation, Building	Zurich, Switzerland
Scene 10	47.630, -122.346	10	265.09	5800	Road, Others	British Columbia, Canada

Cam2BEV [63] addresses the challenge of estimating distances in monocular camera images by transforming the camera perspective into a corrected BEV image. It uses the u-net cross-scale transformer (uNetXST) model to generate a 360° BEV image from multiple vehicle-mounted cameras, making predictions for occluded areas and generalizing well to real-world data using synthetic training data, outperforming traditional IPM. In our experiments, we retained the core architecture of the uNetXST model used in Cam2BEV but adapted the input perspective to the drone's viewpoint.

LSS [11] explicitly predicts the depth distribution from monocular images and uses this information to construct precise 3-D features, enabling better performance in BEV tasks. The model predicts a depth-aware feature map that aligns with the BEV space, allowing for more accurate scene understanding. In our experiments, we adapted LSS by replacing the input perspective with the drone's viewpoint and modified the output head to match the multiclass segmentation structure used in our method.

In our CNN-based model, the ViT [22] structure used in the spatial encoder–decoder of our original network was replaced with a convolutional architecture, while keeping the rest of the pipeline intact, to validate the applicability of our dataset across different model architectures.

The results of the comparison between Cam2BEV, LSS, our convolutional network, and our proposed method are shown in Table III. As demonstrated by Table III, the results not only highlight the effectiveness of our method but also demonstrate the robustness of our dataset in evaluating different approaches across various methods.

C. Visualization

We visualize the classification results on the test set, as shown in Fig. 10. In Fig. 10(a), it can be observed

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TABLE III
COMPARISON OF PERFORMANCE BETWEEN CAM2BEV, LSS, OUR CONVOLUTIONAL NETWORK, AND OUR PROPOSED METHOD. THE MIOU AND F1 SCORES ARE THE AVERAGE VALUES ACROSS SIX CLASSES, INCLUDING "OTHER" CATEGORY

Method	**IoU_vegetation**	**IoU_building**	**IoU_road**	**IoU_ship**	**IoU_water**	**mIoU↑**	**F1↑**
Cam2BEV [64]	0.5338	0.4105	0.5023	0.7368	0.6847	0.6135	0.4886
LSS [11]	0.3514	0.4793	**0.5102**	0.7639	0.5770	0.5518	0.3932
CNN	0.4327	0.4279	0.4413	0.6114	0.7082	0.5736	0.4025
Ours	**0.5931**	**0.6536**	0.5077	**0.8324**	**0.8362**	**0.6973**	**0.5910**

TABLE IV
ABLATION STUDIES OF DIFFERENT TECHNICAL COMPONENTS OF THE PROPOSED METHOD

Name	MPP&Pooling	Encoder1	Encoder2	FPN	IoU_vegetation	IoU_building	IoU_road	IoU_ship	IoU_water	mIoU↑	F1 ↑
Base	✗	✗	✗	✗	0.4175	0.5086	0.3914	0.4110	0.6283	0.4826	0.3475
Model-v1	✗	✓	✓	✓	0.5848	0.4614	0.4482	0.8279	0.8001	0.6293	0.5028
Model-v2	✓	✓	✗	✗	0.4472	0.5839	0.4974	0.3782	0.7658	0.5974	0.4213
Model-v3	✓	✗	✓	✗	0.4381	0.6042	0.4907	0.3044	0.7623	0.6010	0.4918
Model-v4	✓	✓	✓	✗	0.5914	0.6169	**0.5149**	0.5239	0.7739	0.6851	0.5887
Model-v5	✓	✗	✗	✓	0.4420	0.6002	0.3426	0.6817	0.6785	0.6003	0.4876
Ours	✓	✓	✓	✓	**0.5931**	**0.6536**	0.5077	**0.8324**	**0.8014**	**0.6973**	**0.5910**

that even with occlusion and nonuniform scales among input views, the output can still achieve accurate BEV segmentation results at the given resolution. For example, the image depicts partially obscured houses and a swimming pool due to tree cover. In contrast, the CNN architecture model without the transformer encoder fails to extract effective spatial information, leading to significant overfitting issues. Consequently, crucial occlusion details such as the swimming pool are often lost in the side view. In Fig. 10(b), rural roads and some small buildings are well predicted by the model. These are capabilities not possessed by existing methods in the field of autonomous driving. However, in Fig. 10(c), the segmentation performance of our model for densely distributed small boats still needs improvement.

D. Ablation Study

To validate the effectiveness of the three proposed modules in our network, we conducted ablation experiments using a pure convolutional network as the baseline, in which the BEV multiplane projection and spatial encoder–decoder components were directly removed, and the output head was simplified by eliminating the FPN method. Additionally, we developed several variants of the model by incrementally adding different components (Model-v1–Model-v5) to assess their contributions. The results are shown in Table IV.

When the multiplane projection and weighted pooling modules were removed, the network only demonstrated certain perception capabilities for large-area features such as vegetation and water bodies, while its perception capabilities for buildings and roads significantly decreased. This indicates that our BEV modeling strategy effectively projects spatial information, enhancing the perception capabilities for height-sensitive areas (Building) and space-sensitive areas (Road). Incorporating FPN into the decoder resulted in improved performance for categories such as buildings and boats, but a slight decrease in performance for road categories, which we attribute to a trade-off in small-scale target detection and information complexity.

TABLE V
ABLATION STUDIES OF LOSS FUNCTIONS CONFIGURATIONS

L_{ce}	L_{dice} [56]	L_{cls}	mIoU	F1
✓	✗	✗	0.6738	0.5860
✓	✗	✓	0.6804	0.6284
✓	✓	✗	0.6841	0.6403
✓	✓	✓	**0.6973**	**0.5910**

Additionally, we found that adding the class differentiation loss function L_{cls} to the loss function improved detection performance, indicating that this measure helps amplify differences in features between different categories. Experimental results are presented in Table V.

E. Model Robustness Evaluation

In this study, we conducted a series of experiments to evaluate the impact of different numbers of camera views on the performance of multiview collaborative segmentation tasks. First, we trained the network using data from four viewpoints to ensure that the network could fully learn multiview information under optimal conditions. Then, during the testing phase, we performed experiments using three or four viewpoints as the baseline and randomly reduced one viewpoint to simulate scenarios where some drones are occluded or some cameras are unavailable, thereby recording the changes in segmentation performance (such as mean IoU (mIoU) and F1 scores). This process helps us understand the segmentation performance and stability of the network under different numbers of viewpoints.

To further improve the robustness of the network, we introduced a random camera dropout strategy during the training process. In each training batch, we randomly closed one viewpoint or kept the original input to enhance the network's adaptability to different combinations of viewpoints. Subsequently, we employed the same testing method to evaluate the effect of this strategy and verified the improvement in network stability by comparing the experimental results. The experimental results, as shown in Table VI, indicate that the

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TABLE VI
PERFORMANCE METRICS UNDER DIFFERENT TRAINING AND TESTING CAMERA COUNTS AND RANDOM DROPOUT STRATEGY

Training Cameras	Testing Cameras	Random Dropout Strategy	mIoU	F1
4	4	✗	0.6973	0.5910
4	3	✗	0.5942	0.5016
4	3	✓	0.6533	0.5627
3	3	✗	0.6801	0.5840
3	2	✗	0.5707	0.4735
3	2	✓	0.6191	0.5203

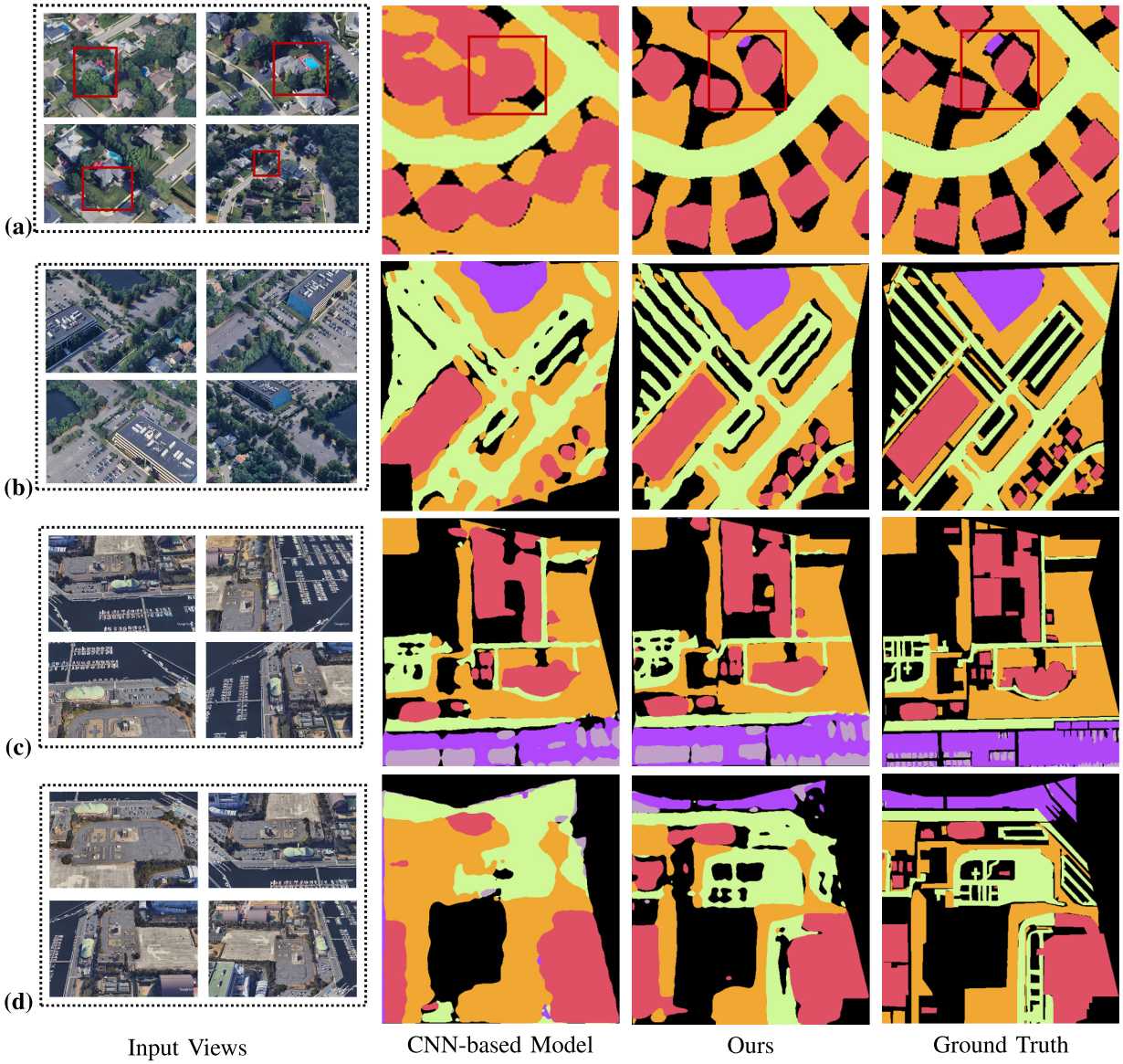


Fig. 10. Visualization of segmentation results between our model and the CNN-based model. (a) Scene with significant occlusions. (b) Scene featuring narrow roads and small buildings. (c) Scene with densely distributed small ships. (d) Complex scene containing multiple elements.

network performance gradually improves with an increasing number of viewpoints. Moreover, by randomly closing some viewpoints during training, the network demonstrated stronger robustness and stability when facing viewpoint loss.

F. Efficiency and Effectiveness

Based on the current model, we obtained four models with different parameter sizes by modifying the dimensions of feature vectors in different modules and the number of layers in the Transformer. For comparative experiments, we also used a pure convolutional architecture with 46 million parameters. Additionally, we tested different models using the same computational resources, and Table VII presents their parameter counts, floating-point operations per second (FLOPs), mIoU, and F1 scores.

We found that our models outperformed the pure convolutional network with the same parameter count (42 million) and

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TABLE VII
PERFORMANCE AND COMPUTATIONAL ANALYSIS OF DIFFERENT MODELS

Name	Backbone	Num_head	Transformer Layer	BEV_dim	Encoder_dim	Params (M)	GFLOPs	mIoU	F1
RSBEV-T	ResNet-18	4	3	32	32	17	108.3	0.4898	0.3103
RSBEV-S	ResNet-50	4	3	32	64	42	167.5	0.6783	0.5629
RSBEV-B	ResNet-50	4	6	64	256	60	184.2	0.6973	0.5910
RSBEV-L	ResNet-50	8	12	256	1024	124	203.4	0.7105	0.5923
CNN-based model	ResNet-50	/	/	64	/	46	116.0	0.5736	0.3425

that performance improved with an increase in the number of Transformer layers.

VI. CONCLUSION

In summary, our work presents RSBEV, a novel approach for multiview collaborative semantic segmentation using BEV representation. Leveraging multiview side-view images under BEV space supervision, our framework improves scene understanding by utilizing 3-D scene information and drone collaborative capabilities. Key contributions include a new task framework that outperforms traditional methods, a novel algorithm with BEV modeling and self-attention mechanisms, and a new dataset for validation. Experiments showed the robustness and adaptability of our model, highlighting the importance of multiplane projection, weighted pooling, and transformer encoder. Evaluation of model robustness, including varying the number of viewpoints and implementing a random camera dropout strategy, demonstrated the network's stability under suboptimal conditions.

This work marks a significant step toward advancing drone-based semantic segmentation and holds promise for applications across various real-world scenarios.

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